

SAI SRESTHA ADUSUMALLI

Sr. DATA ENGINEER

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PROFESSIONAL SUMMARY:

10+ years of extensive experience in Data Engineering, Pipeline Designing, Development and Implementation. My career includes a deep understanding of the Software Development Life Cycle (SDLC), where I've effectively operated within both Waterfall and Agile methodologies to deliver high-quality solutions.

I have a strong experience in Big Data technologies, including core components and ecosystems such as Apache Spark, Spark SQL, Hadoop, HDFS, MapReduce, YARN, Zookeeper, Hive, Pig, HBase, and Sqoop. I specialize in building and sustaining data pipelines using PySpark, BigQuery and Hive, addressing bugs, implementing enhancements and ensuring alignment with business requirements. My expertise also includes writing efficient scripts and APIs in Python, leveraging libraries such as PySpark, pytest, Pymongo, cxOracle, PyExcel, Boto3, Psycpg, NumPy and Beautiful Soup for data analysis and transformation.

An important portion of my work involves cloud computing, where I have hands-on experience with AWS services such as Glue, Redshift, Athena, EMR, S3, RDS and Step Functions. I've also proficiently implemented and managed Azure Data Lakes for data storage and analysis, enhancing the scalability and performance of data solutions. I've also used Google Cloud Platform components including BigQuery, DataProc, DataFlow and Google Cloud Storage. Developed and deployed real-time stream processing applications using Spark, Kafka, Scala and Hive to perform streaming ETL operations. My experience in NoSQL databases like MongoDB and Cassandra has enabled me to design and implement robust distributed data systems. Moreover, I have extended the functionality of Hive and Pig by creating customized User-Defined Functions (UDFs) in Python.

My expertise includes designing and optimizing complex data models both relational and dimensional, using normalization and de-normalization techniques for performance optimization. I have created Slowly Changing Dimensions (SCD) and Fact tables to meet various business needs. I am proficient in SQL across multiple dialects, including MySQL, PostgreSQL, Redshift, SQL Server, and Oracle, and have utilized Teradata utilities like FastExport and MultiLoad for seamless data migration between flat files and source systems. I've significant experience orchestrating workflows and data pipelines using tools like Apache Oozie and Airflow. My proficiency extends to monitoring Hadoop cluster infrastructures using Nagios and Ganglia, ensuring system reliability and efficiency. Apart from that, I've developed MapReduce programs for analyzing large datasets, deployed Spark and Scala code on Hadoop clusters running on GCP and implemented Proof of Concepts (PoCs) and gap analyses to derive actionable insights.

TECHNICALSKILLS:

BigData/Hadoop Technologies	MapReduce,Spark,SparkSQL,Azure,SparkStreaming,Kafka,Pyspark,Pig, Hive, HBase, Flume, Yarn, Oozie, Zookeeper, Hue, Ambari Server
Languages	HTML5,DHTML,WSDL,CSS3,C,XML, R/RStudio,SASEnterpriseGuide, SAS , R (Caret, Weka), Perl, MATLAB, Mathematica, FORTRAN, DTD, Schemas,Json,Ajax,Java,Scala,Python(NumPy,SciPy,Pandas,Genism,Keras), Java Script, Shell Scripting
NOSQL Databases	Cassandra,HBase,MongoDB,MariaDB
WebDesignTools	HTML,CSS,JavaScript,JSP,jQuery,XML
DevelopmentTools	MicrosoftSQLStudio,IntelliJ,AzureDatabricks,Eclipse,NetBeans.
PublicCloud	EC2,IAM,S3,Autoscaling,CloudWatch,Route53,EMR,Redshift

Development Methodologies	Agile/Scrum,UML,DesignPatterns,Waterfall
Build Tools	Jenkins,Toad,SQLLoader,PostgreSQL,Talend,Maven,ANT,RTC,RSA,Control- M, Oozie, Hue, SOAP UI
Reporting Tools	MS Office (Word/Excel/Power Point/ Visio/Outlook), Crystal reports XI, SSRS, Cognos.
Databases	MicrosoftSQLServer2008,2010/2012,MySQL4.x/5.x,Oracle11g,12c,DB2, Teradata, Netezza
Operating Systems	AllversionsofWindows,UNIX,LINUX,MacintoshHD,SunSolaris

PROFESSIONAL EXPERIENCE:

Square, McKinney, Texas

Dec2022 to Present

Senior Data Engineer

Responsibilities:

- Developed Scala and python-based Spark applications for performing data cleansing, event enrichment, data aggregation, de-normalization and data preparation needed for machine learning and reporting teams to consume.
- Worked with Linux systems and RDBMS database on a regular basis to ingest data using Sqoop.
- Created functions and assigned roles in AWS Lambda or run python scripts, and AWS Lambda using Java to perform event driven processing. Created Lambda jobs and configured Roles using AWS CLI.
- Leveraged AWS cloud services such as EC2, auto-scaling and VPC to build secure, highly scalable, and Flexible systems that handled expected and unexpected load bursts.
- Wrote Kafka producer to stream the data from external REST API to Kafka topics.
- Streamed real time data by integrating Kafka with Spark for dynamic price surging using machine learning algorithm.
- Wrote Spark-Streaming application to consume the data from Kafka topics and write the processed stream to S3.
- Used python, the ETL pipeline was developed and programmed to collect data from Redshift data warehouse.
- Used Data bricks for encrypting data using server-side encryption.
- Working on Docker Hub, Docker Swarm, Docker Container network, creating Image Files primarily for middleware installations & domain configurations. Evaluated Kubernetes for Docker Container Orchestration.
- Used Delta Lake as it is an open-sourced data storage layer which delivers reliability to data lakes.
- Experience with Snowflake Multi-Cluster Warehouses.
- Hands on experience moving the data from AWS S3 ecosystem to Snowflake vice versa.
- Worked extensively with Sqoop for importing data from Oracle.
- Involved in creating Hive tables, loading, and analyzing data using hive scripts. Implemented Partitioning, Dynamic Partitions, Buckets in HIVE.
- Creating and writing shell scripts (Bash, Ruby, Python and PowerShell) for automating tasks.
- Worked on Fine-tuning spark applications to improve the overall processing time for the pipelines
- Used Reporting tools like Tableau to connect with Impala for generating daily reports of data.
- Involved in Trouble Shooting, Performance tuning of reports and resolving issues within Tableau Server and Reports.
- Installed Docker Registry for local upload and download of Docker images and from Docker Hub and created Docker Files to automate the process of capturing and using the images.
- Programmatically created CICD Pipelines in Jenkins using Groovy scripts, Jenkins File, integrating a variety of Enterprise tools and Testing Frameworks into Jenkins for fully automated pipelines to move code from Dev Workstations to all the way to Prod environment.
- Worked on troubleshooting spark application to make them more error tolerant.

- Designed, documented operational problems by following standards and procedures using JIRA.
- Responsible for ingesting large volumes of user behavioral data and customer profile data to Analytics Datastore.
- Developed Python scripts to computerize the way towards extricating weblogs utilizing Airflow DAGs.

Customers Bank, Wilmington, North Carolina

Apr 2019 to Nov 2022

Senior Data Engineer

Responsibilities:

- Worked on Big data on AWS cloud services i.e. EC2, S3, EMR and DynamoDB
- Migrated on-premise database structure to Redshift data warehouse
- Was responsible for ETL and data validation using SQL Server Integration Services.
- Defined and deployed monitoring, metrics, and logging systems on AWS.
- Connected to Amazon Redshift through Tableau to extract live data for real-time analysis.
- Used HiveSQL, PrestoSQL and SparkSQL for ETL jobs and using the right technology for the job to get done.
- Created Entity Relationship Diagrams (ERD), Functional diagrams, Data Flow diagrams and enforced referential integrity constraints and created logical and physical models using Erwin.
- Measured Efficiency of Hadoop/Hive environment ensuring SLA is met
- Created ad-hoc queries and reports to support business decisions SQL Server Reporting Services (SSRS).
- Defined facts, dimensions and designed the data marts using the Ralph Kimball's Dimensional Data Mart modeling methodology using Erwin.
- Worked publishing interactive data visualizations dashboards, reports /workbooks on Tableau and SAS Visual Analytics.
- Advanced knowledge on Confidential Redshift and MPP database concepts.
- Designing and building multi-terabyte, full end-to-end Data Warehouse infrastructure from the ground up on Confidential Redshift for large scale data handling Millions of records every day
- Wrote build/integration/installation scripts in python and bash as needed.
- Optimizing and tuning the Redshift environment, enabling queries to perform up to 100x faster for Tableau and SAS Visual Analytics
- Exception handling in python to add logs to the application.
- Analyzed the system for new enhancements/functionalities and perform Impact analysis of the application for implementing ETL changes.
- Migrated ETL solutions from Redshift to run on snowflake database.
- Hands on working experience with snowflake in implementing the row level security.
- Managed security group on AWS, focusing on high-availability, fault-tolerance, and auto scaling using Terraform templates. Along with Continuous Integration and Continuous Deployment with AWS Lambda and AWS code pipeline.
- Compiled data from various sources to perform complex analysis for actionable results
- Built performant, scalable ETL processes to load, cleanse and validate data
- Analyzed the existing application programs and tune SQL queries using execution plan, query analyzer, SQL Profiler and database engine tuning advisor to enhance performance.
- Wrote various data normalization jobs for new data ingested into Redshift.
- Collaborated with team members and stakeholders in design and development of data environment
- Preparing associated documentation for specifications, requirements, and testing
- Developed Unix shell scripts to load large number of files into HDFS from Linux File System
- Participated in the full software development lifecycle with requirements, solution design, development, QA implementation, and product support using Scrum and other Agile methodologies.

Citadel, Houston, Texas

Sep 2017 to Mar 2019

Big Data Engineer

Responsibilities:

- Transforming business problems into Big Data solutions and define Big Data strategy and Roadmap.

Installing, configuring, and maintaining Data Pipelines

- Developed the features, scenarios, step definitions for BDD (Behavior Driven Development) and TDD (Test Driven Development) using Cucumber, Gherkin and Ruby.
- Designing the business requirement collection approach based on the project scope and SDLC methodology.
- Implemented large Lambda architectures using Azure Data platform capabilities like Azure Data Lake, Azure Data Factory, HDInsight, Azure SQL Server, Azure ML and Power BI
- Files extracted from Hadoop and dropped on a daily hourly basis into S3. Working with Data governance and Data quality to design various models and processes.
- Involved in all the steps and scope of the project referenced data approach to MDM, have created a Data Dictionary and Mapping from Sources to the Target in MDM Data Model.
- Involved using ETL tool Informatica to populate the database, data transformation from the old database to the new database using Oracle and SQL Server.
- Designed cloud-based solutions in Azure by creating Azure SQL database, setting up Elastic pool jobs and design tabular models in Azure analysis services.
- Storing Data Files in Google Cloud using DataProc, BigQuery to develop and maintain GCP cloud based solution.
- Working on tickets opened by users regarding various incidents, requests.
- Created a Lambda Deployment function, and configured it to receive events from S3 buckets
- Writing UNIX shell scripts to automate the jobs and scheduling cron jobs for job automation using commands with Crontab.
- Decommissioning nodes and adding nodes in the clusters for maintenance
- Monitored cluster health by setting up alerts using Nagios and Ganglia.
- Adding new users and groups of users as per the requests from the client.
- Developed various Mappings with the collection of all Sources, Targets, and Transformations using Informatica Designer
- Developed Mappings using Transformations like Expression, Filter, Joiner and Lookups for better data messaging and to migrate clean and consistent data
- Used Apache Spark Dataframes, Spark-SQL, Spark MLLib extensively and developing and designing POC's using Scala, Spark SQL and MLLib libraries.
- Creating Pipelines in ADF using Linked Services/Datasets/Pipeline/ to Extract, Transform, and load data from different sources like Azure SQL, Blob storage, Azure SQL Data warehouse, write-back tool and backwards.
- Data Integration ingests, transforms, and integrates structured data and delivers data to a scalable data warehouse platform using traditional ETL (Extract, Transform, Load) tools and methodologies to collect data from various sources into a single data warehouse.
- Applied various machine learning algorithms and statistical modeling like decision trees, text analytics, natural language processing (NLP), supervised and unsupervised, regression models, social network analysis, neural networks, deep learning, SVM, clustering to identify Volume using scikit-learn package in Python, R, and Matlab.
- Create and publish multiple dashboards and reports using Tableau server and work on Text Analytics, Naive Bayes, Sentiment analysis, creating word clouds and retrieving data from Twitter and other social networking platforms.
- Created and maintained SQL Server scheduled jobs, executing stored procedures for the purpose of extracting data from Oracle into SQL Server. Extensively used Tableau for customer marketing data visualization.
- Worked on Apache Flink to implement the transformation on data streams for Filtering, aggregating, update state.
- Used Flink Streaming for pipelined Flink engine to process data streams to deploy new API including definition of Flexible windows.
- Work on data that was a combination of unstructured and structured data from multiple sources and automate the cleaning using Python scripts.

Syntek Solutions, Hyderabad, India

June 2014 to July 2017

Data Engineer

Responsibilities:

- Experience creating and organizing HDFS over a staging area.
- Imported legacy data from SQL Server and Teradata into Amazon S3.
- As a part of Data Migration, wrote many SQL scripts for mismatch of data and worked on loading the history data from Teradata SQL to snowflake.
- Wrote Python code to manipulate and organize data frame such that all attributes in each Field were formatted identically.
- Designed and developed end-to-end ETL processing from Oracle to AWS using Amazon S3, EMR, and Spark.
- Created functions and assigned roles in AWS Lambda to run python scripts, and AWS Lambda using java to perform event driven processing. Created Lambda jobs and configured Roles using AWS CLI.
- Utilized Pandas to create a data frame.
- Implemented UNIX scripts to define the use case workflow and to process the data files and automate the jobs.
- Created bashrc files and all other xml configuration to automate the deployment of Hadoop.
- Developed a raw layer of external tables within S3 containing copied data from HDFS.
- Created a data service layer of internal tables in Hive for data manipulation and organization.
- Developing and maintaining ETL workflows using Apache Nifi, Informatica, or Talend to extract data from various sources, transform it, and load it into data warehouses.
- Designing and managing schemas in Amazon Redshift, Google BigQuery, or SQL Server for efficient data storage and querying.
- Managing databases like MySQL, PostgreSQL, and Oracle Database 12c for transactional and analytical data processing.
- Building and optimizing data pipelines using Apache Hadoop, Hive, and Pig for batch processing.
- Setting up streaming data pipelines with Apache Kafka and Apache Storm to handle real-time analytics.
- Integrating data from APIs, web scraping, and third-party sources using custom scripts in Python or Java.
- Query optimization and tuning in SQL and big data platforms for improved performance.
- Using Git for version control of codebases and configuration files.
- Setting up data backup and recovery processes using tools like AWS S3 and HDFS snapshots.
- Monitoring data pipelines and infrastructure using Nagios, ELK Stack (Elasticsearch, Logstash, Kibana), and Ganglia.
- Writing scripts in Bash, Perl, or Python for task automation and job scheduling.
- Maintaining detailed documentation of ETL processes, data models, and infrastructure configurations.